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“With 5G DEVELOPMENTS Gathering Pace, The Move To An OpenRAN ARCHITECTURE May Be Even MORE URGENT”

OpenRAN solution will provide opportunities for more vendors to participate in the market, and will allow them to drive technical innovations. Marie Ma, senior director - technical marketing solutions and general manager - enterprise business, Comba Telecom International, discusses the components, benefits, risks and future of OpenRAN technology with Deepshikha Shukla of *Electronics For You*

Q. What is virtualisation of radio access network (RAN)?

A. Virtualisation is the ability to simulate a hardware platform in software. All the functionality is separated from the hardware and simulated as a virtual instance, with the ability to operate just like traditional hardware.

Separation of software from hardware makes RAN flexible and helps operators deliver customised services more cost-efficiently.

A virtual RAN works on the principles of network functions virtualisation (NFV). A typical virtual RAN contains a centralised pool of baseband units (BBUs), control functions and service delivery optimisation. With this, BBUs are moved away from the base station to a data centre. As a result, their functions can be implemented with virtual machines in a centralised data centre. This provides intelligent scaling of computing resources while decreasing energy consumption and capex.

Q. What is 5G OpenRAN?

A. OpenRAN is an open ecosystem of general-purpose processing (GPP)-based RAN, which decouples software from hardware vendors and takes advantage of the latest capabilities of GPP platforms, both at the software level and by using programmable offload mechanisms such as FPGAs.

5G OpenRAN is a 5G solution that uses OpenRAN architecture. As OpenRAN is also a virtualised RAN solution, it can deliver most 5G features, like network slicing, NFV, edge computing and massive MIMO, at a much lower cost and greater flexibility. With 5G developments gathering pace, the move to an OpenRAN architecture may be even more urgent.

5G OpenRAN helps operators lower their capex and opex, as it allows to integrate, deploy and operate RAN by using components, sub-systems and software from multiple suppliers connected over open interfaces.

Q. How will virtual Evolved Packet Core (vEPC) and cloud-RAN (C-RAN) change the network dynamics for operators?

A. vEPC is a low-cost virtualised core network solution. A typical vEPC running on a standard server like Intel X86 could support any size and scale of networks efficiently. With vEPC and C-RAN—which can also run on Intel x86—operators can customise networks by mixing and matching individual network components as needed to meet the unique requirements of capacity and other network resources from customers.

vEPC and C-RAN can also lower capex and opex—as these are less reliant on specialised hardware—while speeding service delivery and allowing on-demand scalability, as well as responding to real-time network conditions and user needs.

Q. Are there any concerns related to open source networks?

A. One of the main concerns is awareness about their security. People think proprietary software is inherently more secure than open source software. An open source network does not mean its software and hardware information will be transparent to non-related parties. Regular steps need to be taken to ensure security as in any network, proprietary or otherwise.

Another concern is interoperability of different BBUs and white boxes, especially for large-scale networks.

Q. Are you a member of OpenRAN Alliance or Telecom Infra Project (TIP)?

A. Yes, Comba is a member of both!

Q. How will OpenRAN affect the existing telecom market?

A. It will provide operators with reduced cost and more flexibility, plus more 4G-plus-5G-friendly solutions. It will also allow more vendors to participate.

Q. How are you planning to contribute to its successful deployment?

A. Comba will contribute to the successful commercialisation of OpenRAN with a comprehensive radio solution.

Comba will also work with its OpenRAN partners and bring more innovations to the future network. EFY